

Subject Description Form

Subject Code	AMA502
Subject Title	Operations Research Methods
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Exclusion: MGT 532 "Deterministic Operations Research" or its equivalent.
Objectives	To enable students to appreciate the use of mathematical, computational and statistical techniques in solving real engineering management problems.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: (a) Integrate the mathematical and statistical knowledge and techniques required in operations research model formulation. (b) Execute and appraise the main algorithms for solving operations research problems. (c) Interpret the results of these operations research algorithms. (d) Evaluate critically for improvement in solutions.
Subject Synopsis/ Indicative Syllabus	1. Project management: Networks -- cost analysis of a project; PERT/CPM (Program Evaluation and Review Technique/Critical Path Method); probabilistic considerations of project duration; resources scheduling. 2. Inventory management: EOQ formula -- planned shortage, quantity discounts, and stochastic models. 3. Resource management: Linear programming problems -- formulation of linear programs; dual problems and applications in robust optimization; simplex method; transportation and assignment model. Integer programming problems -- formulation of linear integer programs; branch-and-bound method; cutting plane method.
Teaching/Learning Methodology	The subject will be delivered mainly through lectures and tutorials. The teaching and learning approach is mainly problem-solving oriented. The approach aims at the development of mathematical techniques and how the techniques can be applied to solving problems. Students are encouraged to adopt a deep study approach by employing high level cognitive strategies, such as critical and evaluative thinking, relating, integrating and applying theories to practice.

Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)			
			a	b	c	d
	1. Assignments	10%	✓	✓	✓	✓
	2. Mid-term test	30%	✓	✓	✓	✓
	3. Examination	60%	✓	✓	✓	✓
	Total	100 %				
	Continuous Assessment comprises of assignments/ project and a mid-term test. A written examination is held at the end of the semester.					
Student Study Effort Required	Class contact:					
	▪ Lecture				26 Hrs.	
	▪ Tutorial				13 Hrs.	
	Other student study effort:					
	▪ Assignment				20 Hrs.	
	▪ Case study/Mini-project				38 Hrs.	
	▪ Self-study				40 Hrs.	
	Total student study effort				137 Hrs.	
Reading List and References	DeGroot, Morris H. and Schervish, Mark J.	Probability and Statistics, 5 th Edition	Pearson, 2019			
	Hillier, F.S. and Lieberman, G.J.	Introduction to Operations Research, 10 th Edition	McGraw Hill, 2014			
	Taha, H.A.	Operations Research, 5th Edition	MacMillan, 1992			
	Winston, W. L. and Venkataramanan, M.	Introduction to Mathematical Programming, Operations Research: Volume One, 4 th Edition	Brooks/Cole-Thomson Learning, 2002			
	Ahuja, R. Magnanti, T. and Orlin, J.	Network Flows: Theory, Algorithms and Applications, 1 st Edition	Pearson, 1993			
	Wolsey, A. L., and Nemhauser, G. L.	Integer and Combinatorial Optimization	Wiley, 1999			